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Weakly Supervised Semantic Segmentation via Progressive Confidence Region Expansion

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Abstract

Weakly supervised semantic segmentation (WSSS) has garnered considerable attention due to its effective reduction of annotation costs. Most approaches utilize Class Activation Maps (CAM) to produce pseudo-labels, thereby localizing target regions using only image-level annotations. However, the prevalent methods relying on vision transformers (ViT) encounter an "over-expansion" issue, i.e.*, CAM incorrectly expands high activation value from the target object to the background regions, as it is difficult to learn pixel-level local intrinsic inductive bias in ViT from weak supervisions. To solve this problem, we propose a Progressive Confidence Region Expansion (PCRE) framework for WSSS, it gradually learns a faithful mask over the target region and utilizes this mask to correct the confusion in CAM. PCRE has two key components: Confidence Region Mask Expansion (CRME) and Class-Prototype Enhancement (CPE). CRME progressively expands the mask in the small region with the highest confidence, eventually encompassing the entire target, thereby avoiding unintended coverage of background areas. CPE aims to enhance mask generation in CRME by leveraging the similarity between the learned, dataset-level class prototypes and patch features as supervision to optimize the mask output from CRME. Extensive experiments demonstrate that our method outperforms the existing single-stage and multi-stage approaches on the PASCAL VOC and MS COCO benchmark. Our code is available at https://github.com/xxf011/WSSS-PCRE.*

1. Introduction

Weakly supervised semantic segmentation (WSSS), which predicts precise pixel-level object masks using weak annotations, has received a great deal of research focus in recent years. These weak annotations include image-level labels [1, 2], points [3], scribbles [4, 5], and bounding boxes [6, 7]. Among them, image-level annotation is one of the most attractive and challenging options due to its easy acquisition with the least amount of information. This work also focuses on WSSS with image-level labels, aiming to explore how to achieve high-quality pixel-level predictions.

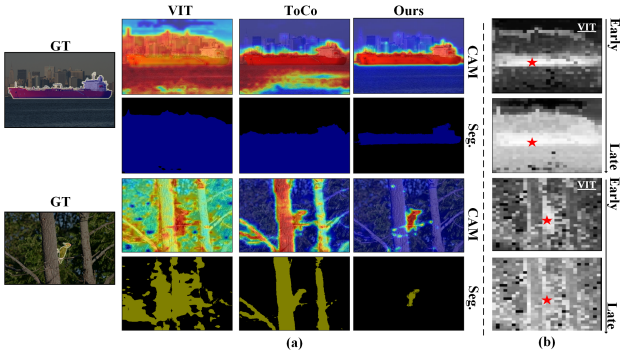


Figure 1. **(a) Comparison of generated CAM and segmentation maps between previous methods and ours.** Our PCRE approach effectively addresses incorrect segmentation caused by over-expanded activations propagating from the target object region. 'Seg.' denotes segmentation maps. **(b) Cosine similarity between the patch with a red star (★) and other patches.** Object regions and backgrounds have high similarities (white color) in the late training process.

Prevalent WSSS methods [2, 8–10] typically employ Class Activation Maps (CAMs) [11] to extract localization maps as pseudo labels. These methods adopt multi-stage pipelines: generating initial pseudo labels via CAMs with CNNs, refining them, and finally training fully supervised segmentation models with these refined labels. However, the complicating multi-stage training slows down the efficiency. Therefore, many studies [10, 12–14] have proposed a one-stage training way to produce pseudo labels and learn a segmentation head simultaneously. Meanwhile, CNNs fail to effectively model the global feature relations, which hinders the activation of the entire object regions. Recently, WSSS methods [15] adapt vision transformer (VIT) architectures to capture long-range relationships with

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self-attention blocks, improving the accuracy of pseudo labels [12, 13, 16]. For example, Rossetti *et al.* [16] leverages global max pooling into the ViT features in the last layer to optimizing CAMs and ToCo [12] employs contrastive learning to extract local semantic information from image patches.

Despite these advancements, a significant challenge remains that these methods tend to incorrectly expand CAM from the target object region to the background with high activation values. We call this phenomenon as *over-expansion* of CAM. As shown in Fig. 1 (a), the target bird region easily spreads to the adjacent tree trunk (*i.e.*, background) in the generated CAM after training. This is due to that the weak annotations fail to learn inductive biases in modeling local feature structures, which is lacking in ViT, and high feature similarities between the target and background result in unreliable pseudo labels of background. In Fig. 1 (b), as training proceeds, the high feature similarity expands from the target inferior to its exterior background, further intensifying the over-expansion in CAM.

To solve the above issue, we propose a novel weakly supervised semantic segmentation framework via *Progressive Confidence Region Expansion* (PCRE), which gradually generates an accuracy and complete target mask for effective WSSS. As illustrated in Fig. 2, the proposed PCRE consists of two modules, Confidence Region Mask Expansion (CRME) and Class-Prototype Enhancement (CPE). Specifically, CRME incrementally expands the scope of the segmentation mask in small steps to constrain the mask to diffuse background areas. It begins by selecting a mask over the region with the highest confidence. After that, it restricts the mask generation range at each expansion step to progressively incorporate additional masks based on their patch confidence levels, which produces an accurate refined mask over the entire target. However, relying solely on the CRME mask expansion strategy still struggles to fully capture the semantic precision of target regions. To address this, we propose CPE to further guide the CRME expansion process. We observe that shallow layers in the early stages can capture discriminative local features relevant to the target, but these features tend to be overly smoothed under the global attention mechanism as training progresses. As such, CPE introduces the EMA to preserve these early-stage discriminative features, which optimizes dense similarity maps through class prototype alignment to capture fine-grained semantic correlations across images.

Our main contributions can be summarized as:

- We propose Progressive Confidence Region Expansion (PCRE) to address the over-expansion issue for effective WSSS. By progressively expanding high-confidence regions, PCRE significantly improves CAM pseudo labels and segmentation performance.
- We design a CRME module that incrementally expands the mask's scope in small steps, effectively avoiding unintended background coverage. Additionally, we develop a CPE module to learn rich, dataset-level representations for each class, generating the prototypical CAM for effective supervision.
- Experimental results demonstrate that the superior performance of our method. For example, on the COCO 2014 benchmark dataset [17], the proposed PCRE outperforms SOTA single-stage DIAL [14] and multi-stage PCSS [18] by 2.8% mIoU and 1.5%, respectively.

2. Related work

Weakly-Supervised Semantic Segmentation. Traditional WSSS [2, 8] primarily relies on CNN to localize target regions through CAM [11]. However, CAM typically only covers the most distinctive parts of the objects [19, 20]. To address this limitation, researchers have proposed methods such as adversarial erasing [21–23], auxiliary tasks [24–26], and CRF [27] to expand activation regions and improve the quality of pseudo labels. With the emergence of ViT [15], researchers have explored the application of ViT in WSSS, leveraging the global semantic aggregation capability of the class token to enhance the model's comprehensive focus on target regions. For example, the works in [28, 29] introduce multiple class-specific tokens and utilized cross-attention maps between class tokens and patch tokens to acquire positional cues. the works in [12, 30] contrast class tokens from full-image labels with those from cropped uncertain regions. DIAL [14] incorporates cross-modal contrastive learning to maximize similarity between class tokens and target text embeddings. CTI [31] enhances the consistency and class specificity of CAM by injecting class tokens across different augmentations of the same image and between different images. However, these methods either employ a single class token for the entire image, limiting effective class differentiation in multi-class scenarios, or introduce multiple class tokens with cross-attention mechanisms that introduce noise due to their indirect association with the optimization objective. We propose the CPE module to extract both nuanced semantic information and representative, dataset-level prototypes for each class.

Progressive Expansion. In the computer vision literature, the concept of progressive expansion has been explored in various tasks, but with motivations and implementations very different from our approach. For instance, PDEN [32] progressively expands from single to multiple domains to simulate various photometric and geometric transforms in unseen domains. PSS [33] progressively enlarges the training set to enhance model resilience against out-of-distribution classes. BungeeNeRF [34] incrementally expands the training dataset, enabling the model to robustly capture radiance representations across varying scales within a scene. ToE [35] progressively expands to-

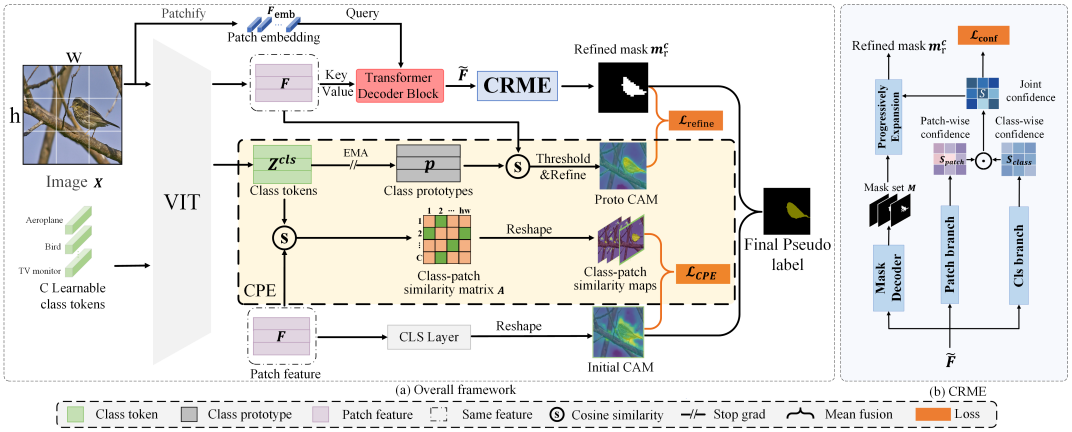


Figure 2. **(a) The overall framework of the proposed PRCE.** It comprises two inputs: an image and C learnable class tokens. Initially, a CLS layer is employed to generate an initial CAM. CRME incrementally broadens the segmentation mask's scope based on each patch's confidence, ultimately producing a refined mask over the entire target region. CPE first utilizes the initial CAM to guide the similarity between the learnable class tokens and patch features, ensuring the derivation of representative class tokens. Subsequently, it applies an EMA to create robust, dataset-level prototypes for each class. The similarity between patch features and these prototypes serves as a pseudo-label, optimizing the refined mask to improve the effectiveness of CRME. Finally, the final CAM is produced by taking the average of the refined mask and the initial CAM. **(b) An illustration of CRME.** It comprises three branches, which leverages the joint confidence from patch and class brach into the mask expansion to generate the refined mask.

kens from a limited set to a full set during training to achieve training acceleration. Different from previous works, our method leverages progressive expansion based on each patch's confidence to reduce model confusion when handling difficult-to-segment regions in images, such that mitigating the risk of misclassifying these regions into the same class in CAM.

3. Methods

3.1. Preliminaries

We define X as an input image and $\mathbf{y}$ as its corresponding class label, where $\mathbf{y} \in \{0,1\}^C$ indicates the one-hot vector with the total C classes. In recent WSSS methods [13, 14], initial pseudo-labels are usually generated from the features extracted from ViT by CAM [11]. specifically, the image feature map $F \in \mathbb{R}^{hw \times d}$ is extracted by the backbone, where h, w, and d represent the height, width, and dimension of the feature map, respectively. Note that hw also presents the total token number in ViT. Subsequently, the image feature F is multiplied by the weights $W^{\text{cls}} \in \mathbb{R}^{C \times d}$ of the classification layer to generate initial $\text{CAM}_{\text{init}} \in \mathbb{R}^{C \times hw}$, which is formulated as:

$$\text{CAM}_{\text{init}} = \text{Norm}(\text{ReLU}(W^{\text{cls}} F^{\top})), \tag{1}$$

where $\text{Norm}(\cdot)$ denotes the min-max normalization by scaling CAM to the range of $[0,1]$. Generally, the pseudo labels are generated by a threshold β to differentiate between the foreground and background regions.

3.2. Overview

As presented in Fig. 2, the proposed PCRE aims to reduce the over-expansion problem, which comprises two key components: Confidence Region Mask Expansion (CRME) and Class-Prototype Enhancement (CPE). CRME dynamically generates masks from small high-confidence regions, which progressively expand the refined masks m_r^c over the entire target region. CPE employs exponential moving average (EMA) to dynamically optimize class tokens Z^{cls}, integrating features from different stages to generate representative class prototypes p. The prototypical CAM_{pro} between class prototypes p and image features F serves as the supervision for the refined masks m_r^c in CRME. Finally, the pseudo labels are obtained by combining the prototypical CAM_{pro} and the initial CAM_{init}, which generates the pseudo-labels to supervise the final segmentation branch.

3.3. Confidence Region Mask Expansion

The core idea of CRME is to gradually expand the mask's scope in small increments based on the confidence level of each patch, to effectively restrict the mask's spread into the background regions.

Patch-level Confidence Allocation. To assign a faithful confidence level for each patch, we introduce Patch-level Confidence Allocation to improve its confidence score. We first patchify input image X to obtain embedding features F_{emb}. Then, the final layer output features F are extracted from ViT, and two transformer layers are applied to generate the merged features $\tilde{F}$ by taking the inputs from F_{emb} as the query and F as the key and value. The previous methods [12] usually apply a classification layer to $\tilde{F}$ to

compute the patch-level classification scores. However, it is difficult to capture the relative confidence between classes and patches. Inspired by [36], we introduce two additional branches, *i.e.*, patch branch and class branch, to model both class-specific information and spatial relations. Those branches process the merged features $\tilde{F}$ along the class and patch dimensions, which are formulated as:

$$S^{i,j}_{\text{class}} = \frac{e^{(W^i_{\text{class}}\tilde{F}^j)}}{\sum_{l=1}^{C} e^{(W^l_{\text{class}}\tilde{F}^j)}}, \quad S^{i,j}_{\text{patch}} = \frac{e^{(W^i_{\text{patch}}\tilde{F}^j)}}{\sum_{k=1}^{hw} e^{(W^i_{\text{patch}}\tilde{F}^k)}}, \quad (2)$$

where $S^{i,j}_{\text{class}}$ denotes the confidence score for patch i concerning class j, $S^{i,j}_{\text{patch}}$ represents the confidence score for class i across patch j, $W_{\text{class}}, W_{\text{patch}} \in \mathbb{R}^{C\times d}$ are weight matrices for the class dimension with W^l_{class} and W^i_{patch} denoting the weights for class l and class i respectively, $\tilde{F}^i \in \mathbb{R}^{d\times 1}$ represents the feature of patch i, while C and hw represent the total number of classes and patches respectively. We then derive the joint confidence score by taking the Hadamard product of the class-specific and patch-specific confidence scores:

$$S = S_{\text{class}} \odot S_{\text{patch}}, \quad (3)$$

where $S \in \mathbb{R}^{C\times hw}$ is the joint confidence score, $\odot$ denotes Hadamard product. Therefore, we can construct the confidence loss $\mathcal{L}_{\text{conf}}$ by using class label y as the supervision to optimize S:

$$\mathcal{L}_{\text{conf}} = \frac{1}{C}\sum_{c=1}^{C} \texttt{BCE}\left(\sum_{i=1}^{hw} S_{c,i}, \mathbf{y}_c\right), \quad (4)$$

where $\texttt{BCE}(\cdot,\cdot)$ is the binary cross-entropy function. This allocation method generates reliable S, enabling the extraction of semantically consistent, high-confidence regions, to guide the Patch-level mask generation.

Patch-level Mask Generation. Additionally, we design a mask generation branch that adopts a lightweight MaskFormer segmentation decoder [37] structure to generate a mask set $M = \{m_{i,j} \mid i = 1,\ldots,h;\ j = 1,\ldots,w\}$, with each $m_{i,j} \in [0,1]^{h\times w}$ corresponding to the mask for the patch located at position (i,j), *i.e.* each patch has a mask of size $h \times w$. The mask decoder dynamically adjusts the generated mask based on each patch's features, ensuring that the semantic information in the mask is closely related to the patch.

We further redefine each mask $m_{i,j}$ by restricting path predictions as follows:

$$m_{i,j}(x,y) = \begin{cases} m_{i,j}(x,y) & \text{if } (x-i)^2 + (y-j)^2 \le r^2 \\ 0 & \text{otherwise} \end{cases}, \quad (5)$$

where each patch only predicts values within nearby regions. In this formulation, $m_{i,j}$ retains its original values within a circular region centered at (i,j) with radius r, while values outside this region are set to zero. This constraint ensures that each patch only contributes predictions within a defined local area, thereby limiting its receptive field to its immediate neighborhood. Meanwhile, the radius r gradually increases during training. To facilitate subsequent descriptions, we express the mask set M in its flattened form as $M = \{m_k \mid m_k = m_{i,j},\ k = i\cdot w + j\}$, where k denotes the flattened index.

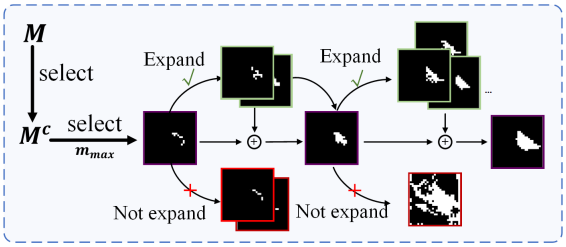


Figure 3. **Visualization of Progressive Mask Expansion and Combination.** The arrows indicate steps of mask expansion, and $\oplus$ represents the operation of adding a mask to the set $\hat{M}^c$. For visual clarity, the masks in $\hat{M}^c$ are shown overlaid, and each mask is displayed in its binarized form.

Progressive Mask Expansion and Combination. Supervising the mask generation with CAM is insufficient to further enhance the segmentation performance. Therefore, we propose a progressive mask expansion and combination strategy, *i.e.*, starting from the region with the highest confidence and gradually extending to the complete region of the target. In the initial stage of the strategy, for class c, the mask with the highest confidence score is selected by applying $\arg\max$ to the confidence score S, and the candidate mask set M^c is constructed as:

$$M^c = \left\{ m_i \mid \arg\max_j S(i,j) = c \right\}, \quad (6)$$

where class c has the highest confidence score in m_i, thus forming the candidate mask set for class c. Next, we filter and expand the candidate masks, according to the mask selection, as presented in Alg. 1. These steps utilize the **Reference Overlap Ratio (ROR)** metric, which measures the overlap ratio relative to a reference mask. The ROR metric is defined as the ratio of the intersection area between two masks to the area of the reference mask:

$$\text{ROR}(m_i, m_j) = \frac{|\mathbb{I}(m_i \ge \tau_{\text{bin}}) \cap \mathbb{I}(m_j \ge \tau_{\text{bin}})|}{|\mathbb{I}(m_i \ge \tau_{\text{bin}})|}, \quad (7)$$

where m_i and m_j is the reference mask and the candidate mask, respectively. $\mathbb{I}$ is an element-wise indicator function matrix used to generate a binarized mask based on the threshold τ_{bin}. The expansion process is illustrated in Fig. 3. Initially, masks with a high overlap with the reference, *i.e.*, they ROR greater than or equal to the high threshold τ_{high},

Algorithm 1 Class Mask Selection.

Input: Joint confidence score S^c for class c, candidate masks set $M^c = \{m_1^c, m_2^c, \ldots, m_n^c\}$ for class c, $n = |M^c|$, Reference Overlap Ratio $\text{ROR}(m_i, m_j)$, threshold parameters τ_{high}, τ_{low}.

Output: Combined mask m_{r}^c for class c.

Initialize $\hat{M}^c = \emptyset$
Find the highest scoring mask $m_{\text{max}} \leftarrow m_i^c$, where $i = \arg\max(S^c)$.
Add m_{max} to $\hat{M}^c$, i.e., $\hat{M}^c \leftarrow \hat{M}^c \cup \{m_{\text{max}}\}$.
repeat
  for all $m_j \in M^c$ **do**
    $R^* \leftarrow \max_{m_i \in \hat{M}^c} \text{ROR}(m_i, m_j)$
    if $R^* > \tau_{\text{high}}$ **then**
      $M^c \leftarrow M^c \setminus \{m_j\}$
    else if $\tau_{\text{low}} < R^* \leq \tau_{\text{high}}$ **then**
      $\hat{M}^c \leftarrow \hat{M}^c \cup \{m_j\}$
      $M^c \leftarrow M^c \setminus \{m_j\}$
    end if
  end for
until $\hat{M}^c$ no longer changes
$m_{\text{r}}^c \leftarrow \text{Max}(\text{Stack}(\hat{M}^c), \text{dim} = 0)$
Return m_{r}^c

are removed from the candidate set M^c to avoid redundancy and to prevent large masks from excessively covering the selected masks (like the red box in the lower right corner mask in Fig. 3). Thus, it avoids reckless spread into background regions. The algorithm then identifies masks with a moderate overlap, *i.e.*, ROR between the low threshold τ_{low} and the high threshold τ_{high} from the remaining unselected masks and adds them to the selected mask set $\hat{M}^c$, thereby progressively expanding the mask coverage area.

This process iterates until the selected mask set $\hat{M}^c$ stabilizes and no further changes occur. Through these iterations, the selected mask set gradually expands from regions with high confidence to encompass the entire target area. The mask m_r^c is generated by stacking all selected masks together to produce the complete segmentation output. For each class, this operation is repeated T times to ensure coverage of multiple instances and high-confidence regions, after which the resulting m_r^c masks are also stacked to form a final refined mask m_r^c. As such, the refined mask loss $\mathcal{L}_{\text{refine}}$ is constructed as:

$$\mathcal{L}_{\text{refine}} = \frac{1}{|\{c \mid \mathbf{y}_c \neq 0\}|} \sum_{\{c|\mathbf{y}_c \neq 0\}} \text{BCE}(m_r^c, Y_{\text{pro}}^c), \tag{8}$$

where $\{c \mid y^c \neq 0\}$ denotes the set of present classes, and Y_{pro} is the enhanced prototype pseudo label generated from the CPE module, which will be described in Sec. 3.4. Therefore, the CRME loss is defined as:

$$\mathcal{L}_{\text{CRME}} = \lambda_1 \mathcal{L}_{\text{conf}} + \lambda_2 \mathcal{L}_{\text{refine}}, \tag{9}$$

where λ_1 and λ_2 are weighting parameters for each loss term.

3.4. Class-Prototype Enhancement

CPE aims to learn discriminative, dataset-level prototypes for each class, which is further utilized in the patch features to generate high-quality pseudo labels, serving as supervision for mask generation in CRME.

Class-Specific Semantic Alignment. This module aims to provide each CLS token with more distinct semantic information by supervising the alignment between CLS tokens and image features. In the vanilla ViT, the CLS token $z^{\text{cls}} \in \mathbb{R}^d$ is projected into classification logits through a linear layer and trained using cross-entropy loss with label $\mathbf{y}$. It enables efficient aggregation of global features for classification but fails to obtain rich semantic information to present different categories for segmentation [29]. We consider adding additional C learnable class tokens, which generate the class token matrix $Z^{\text{cls}} \in \mathbb{R}^{C \times d}$ via ViT. To further form semantic matches between the CLS tokens and the image features at a spatial level, we propose the similarity map between class tokens and image features to enhance their interaction via the following computation:

$$A = \frac{Z^{\text{cls}} \cdot F^\top}{\|Z^{\text{cls}}\| \cdot \|F\|}, \tag{10}$$

where $A \in \mathbb{R}^{C \times hw}$. Previous studies [12] show that the features from the last layer of the backbone tend to be overly smooth, which may lead to background regions being mistakenly identified as foreground. Therefore, we utilize the CAMs from an intermediate layer to generate pseudo-labels, denoted as $Y_{\text{a}} \in \mathbb{R}^{hw}$, thereby capturing higher-confidence foreground regions and using them to supervise the similarity map A. The supervised loss is defined as:

$$\mathcal{L}_{\text{CPE}} = \frac{1}{N^+} \sum_{i=Y_{\text{a}}^j} (1 - A_{i,j}) + \frac{1}{N^-} \sum_{i \neq Y_{\text{a}}^j} A_{i,j}, \tag{11}$$

where N^+ and N^- are the numbers of positive and negative pairs, respectively. $A_{i,j}$ is the similarity between class i and position j, and Y_{a}^j denotes the pseudo-label for position j. Additionally, we utilize the absolute similarity rather than its original formulation to promote diversity [12, 38].

Progressive Class Prototype Update. We observed that the backbone tends to focus on the most discriminative regions (local information) of the target during the early stages of training, which can be used to alleviate the over-expansion of high feature similarity from the target's interior to its exterior. To this end, we introduce class-specific prototype p_i, which uses early-stage features to enrich the model's generalization capacity across different images. Specifically, each class-specific token Z_i^{cls} has an associated prototype $p_i \in \mathbb{R}^d$, which is updated using Exponential Moving Average (EMA) according to the formula:

$$p_i = \lambda_p p_i + (1 - \lambda_p) Z_i^{\text{cls}}, \tag{12}$$

where λ_p is the smoothing factor controlling the update rate of the prototype. These prototypes accumulate information across images and refine the representation of class tokens by balancing early local discriminative features with global class characteristics. Finally, we compute the refined similarity map (denoted as CAM_{pro}) between the prototype p and the feature map F by using Eq. 10. Thus, the enhanced pseudo label $Y_{\text{pro}} \in \mathbb{R}^{C\times hw}$ can be obtained by thresholding CAM_{pro}, which are then used to optimize the mask generation in CRME. All samples within a category share a single class prototype throughout training, encapsulating discriminative, dataset-level information specific to that category.

3.5. Optimization

The overall loss function for learning pseudo labels is formulated as the weighted sum of the above losses:

$$\mathcal{L}_{\text{pseudo}} = \mathcal{L}_{\text{CRME}} + \lambda_3 \mathcal{L}_{\text{CPE}} + \mathcal{L}_{\text{cls}}, \tag{13}$$

where λ_3 is weighting parameter for $\mathcal{L}_{\text{CPE}}$, and classification losses $\mathcal{L}_{\text{cls}}$ are applied to generate initial CAMs. For each class c, the final pseudo labels are computed via the average combination of the initial CAM and the refined mask m_r^c. We apply the common cross-entropy loss $\mathcal{L}_{\text{seg}}$ to the final segmentation branch. Thus, the total loss for segmentation is constructed as:

$$\mathcal{L} = \mathcal{L}_{\text{pseudo}} + \mathcal{L}_{\text{seg}}. \tag{14}$$

Following previous approaches [10, 12, 13], an additional regularization loss [39] is applied to enhance the spatial consistency of the predicted segmentation masks.

4. Experiments

4.1. Experimental Settings

Datasets and Evaluation Metrics. We conduct experiments on the PASCAL VOC 2012 [40] and MS COCO 2014 [17]. For the VOC 2012, the SBD dataset [41] is integrated into the training set as part of a common data augmentation strategy. The expanded VOC 2012 dataset consists of 10,582 training images, 1,449 validation images, and 1,456 test images, with test set performance evaluated on the official evaluation server. The COCO 2014 dataset contains 82,081 and 40,137 images for training and validation sets, respectively. We use the Intersection over Union (mIoU) metric to assess the experimental results.

Implement Details. We utilize a ViT-B/16 backbone pretrained on ImageNet 21k. Optimization is conducted using the AdamW optimizer with a polynomial learning rate scheduler, where the learning rate for the backbone is set to $3e-5$, and $3e-4$ for other components. During inference, multi-scale testing and dense CRF post-processing [27] are employed to keep consistent with prior works [12, 13]. By default, the background thresholds β is set as 0.5. During patch-level mask generation, the radius r is constrained using cosine updating to optimize local region coverage. The number of iterations T for each class in the mask expansion process is set to 5. In mask expansion, the overlap control thresholds $(\tau_{\text{high}}, \tau_{\text{low}})$ are set to $(0.5, 0.4)$ and τ_{bin} is set to 0.7. Additionally, the EMA factor of $\lambda_p = 0.99$ is applied to stabilize the prototype updates. The loss function weights are set as $(\lambda_1, \lambda_2, \lambda_3) = (2.0, 1.0, 0.02)$. More training details can be referred to Supplementary Materials.

Methods	BackBone	Train	Val
Multi-stage WSSS methods.			
CLIMS [42] CVPR'22	ResNet101	70.5	-
CLIP-ES [43] CVPR'23	ResNet101	70.8	-
PCSS [18] ECCV'24	DeiT-S	73.2	-
CTI [31] CVPR'24	ResNet38	73.7	-
One-stage WSSS methods.			
ToCo [12] CVPR'23	VIT-B/16	73.6	72.3
DIAL [14] ECCV'24	VIT-B/16	75.2	73.1
DuPL [13] CVPR'24	VIT-B/16	76.0	74.1
Ours	**VIT-B/16**	**77.6**	**76.3**

Table 1. **Comparison of CAM pseudo labels.** Evaluation is conducted on the PASCAL VOC 2012 train and val sets.

Methods	**Sup.**	**Backbone**	**VOC**		**COCO**
			Val	Test	Val
Multi-stage WSSS methods.					
CLIMS [42] CVPR'22	I+L	ResNet101	70.4	70.0	-
CLIP-ES [43] CVPR'23	I+L	ResNet101	73.8	73.9	45.4
L2G [24] CVPR'22	I+S	ResNet101	72.1	71.7	44.2
PPC [44] CVPR'22	I+S	ResNet101	72.6	73.6	-
FPR [45] ICCV'23	I	ResNet101	70.3	70.1	43.9
MCTformer [29] CVPR'22	I	ResNet38	71.9	71.6	42.0
OCR [46] CVPR'23	I	ResNet38	72.7	72.0	42.5
PCSS [18]ECCV'24	I	ResNet38	73.2	73.0	45.7
CTI [31] CVPR'24	I	ResNet38	74.1	73.2	45.5
One-stage WSSS methods.					
1Stage [47] CVPR'20	I	ResNet38	62.7	64.3	-
AFA [10] CVPR'22	I	MiT-B1	66.6	66.3	38.9
TSCD [48] AAAI'23	I	MiT-B1	71.1	72.2	40.1
ToCo [12] CVPR'23	I	VIT-B/16	71.1	72.2	42.3
DuPL [13] CVPR'24	I	VIT-B/16	73.3	72.8	44.6
DIAL [14] ECCV'24	I+L	VIT-B/16	74.5	74.9	44.4
Ours	I	**VIT-B/16**	**75.5**	**75.9**[1]	**47.2**

Table 2. **Comparison of the semantic segmentation results.** Semantic segmentation performance is evaluated on the PASCAL VOC validation and test sets, along with the COCO validation set. "Sup." denotes the supervision type: I for image-level labels, S for saliency maps, and L for image-level language annotations.

4.2. Experimental Results

Evaluation of Pseudo Labels. In Tab. 1, we demonstrate the performance advantage of our model on the VOC training set, compared to current state-of-the-art methods

[1] http://host.robots.ox.ac.uk:8080/anonymous/ZALC8B.html

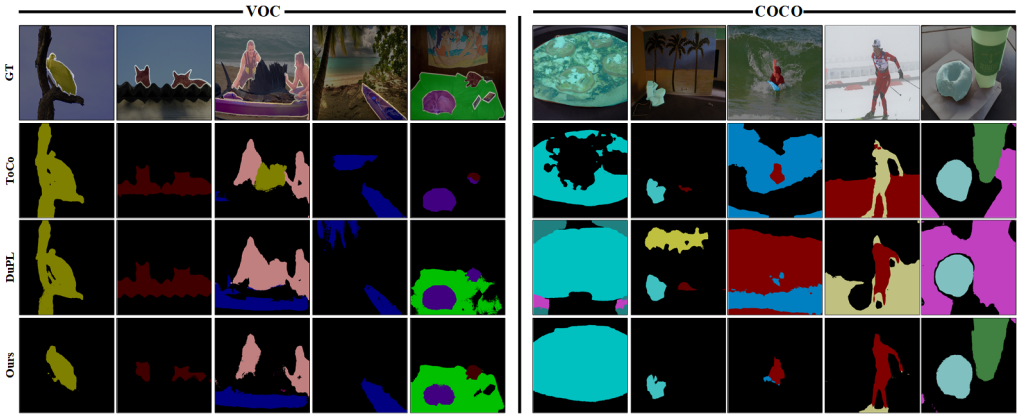


Figure 4. **Visualization results of Segmentation on the PASCAL VOC 2012 and MS COCO datasets.** Our method demonstrates more precise segmentation results.

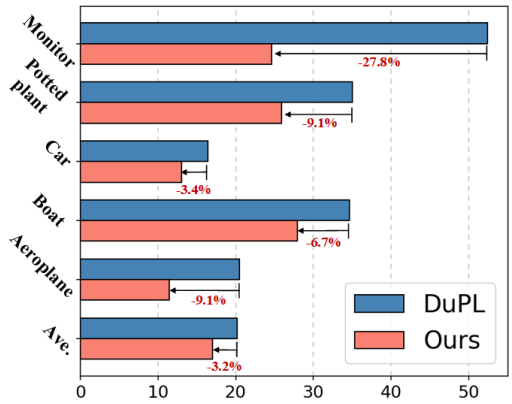


Figure 5. **The OA rate (%) of CAM evaluated on the VOC val set.** Lower OA indicate more accurate segmentation, with reduced mislabeling and fewer unintended expansions beyond target boundaries.

through the evaluation of CAM pseudo labels. With the CRME and PCE, our generated CAM pseudo label achieves 77.6% mIoU in the VOC 2012 train set, surpassing all other methods with multi-stage and one-stage training. In addition, our PCRE also outperforms the one-stage DuPL [13] by 2.2% on the VOC validation set, further demonstrating the effectiveness of our method in generating high-quality pseudo-labels.

Evaluation of Semantic Segmentation. Tab. 2 presents the semantic segmentation performance of our method on the VOC and COCO datasets. Our method achieves 75.5% and 75.9% mIoU on the VOC validation and test sets, which outperforms one-stage DIAL [14] by 1.0% and 1.0% mIoU, respectively. For the evaluation of the COCO validation, our method also achieves new state-of-the-art results with 47.2% mIoU, significantly outperforming DIAL by 2.8% mIoU. Moreover, our method also achieves 1.5% mIoU gains over multi-stage PCSS [18].

Quantitative Evaluation of Over Expansion Reduction. To quantify the effectiveness of our method in alleviating the over-expansion issue, we employ the over-activation rate (OA), as defined in DuPL [13]. OA represents the ratio of background pixels incorrectly labeled as the target, thereby measuring the extent of unintended over-expansion into non-target areas. This is directly aligned with the over-expansion problem, where high OA values indicate severe expansion beyond target boundaries. The OA metric, defined as $\text{OA} = \frac{FP}{TP+FP}$ provides an objective basis for comparing methods on their ability to reduce such target-background confusion. As shown in Fig. 5, our method significantly reduces the OA rate for the monitor by 27.84%, and the average OA rate per category decreases by 3.2% compared to DuPL. These results demonstrate that our method effectively reduces the target confusion by carefully expanding the target region.

Methods	Backbone	Val	Ratio
Fully-supervised methods			
DeepLabV2 [49] TPAMI'17	ResNet101	77.7	100%
WideResNet [50] PR'19	ResNet38	80.8	100%
Segformer [51] ICCV'21	MiT-B1	78.7	100%
DeepLabV2 [49] TPAMI'17	ViT-B/16	82.3	100%
Multi-stage WSSS methods			
AdvCAM [52] CVPR'21	ResNet101	68.1	87.6%
PPL [53] TMM'23	ResNet38	67.8	87.3%
MCTformer [29] CVPR'22	ResNet38	71.9	89.0%
CTI [31] CVPR'24	ResNet38	74.1	91.7%
One-stage WSSS methods			
SLRNet [54] IJCV'22	ResNet38	69.3	85.8%
AFA [10] CVPR'22	MiT-B1	66.0	83.9%
ViT-PCM [16] ECCV'22	ViT-B/16	70.3	85.4%
ToCo [12] CVPR'23	ViT-B/16	71.1	86.4%
DuPL [13] CVPR'24	ViT-B/16	73.3	89.1%
Ours	**ViT-B/16**	**75.5**	**91.7%**

Table 3. **The comparison to the fully-supervised counterparts on VOC val set.** For fairness, this evaluation is restricted to methods with the only usage of image-level labels. The ratio represents the performance ratio between WSSS methods and their fully-supervised ones based on the same backbone.

Baseline	CRME		CPE		CAM	Seg.
	PCA	PMEC	$\mathcal{L}_{\text{CPE}}$	PCPU		
✓					70.7	68.9
✓	✓				72.1	70.5
✓	✓	✓			74.4	72.5
✓	✓	✓	✓		75.5	73.2
✓	✓	✓	✓	✓	**76.3**	**73.8**

Table 4. **Abalation study of key components on VOC val set. PCA**, **PMEC**, $\mathcal{L}_{\text{CPE}}$ and **PCPU** denote Patch-level Confidence Allocation, Progressive Mask Expansion and Combination, CPE loss and Progressive Class Prototype Update, respectively. CRF postprocessing is not conducted in the ablation study.

Comparison by Reducing the Effect of Backbones. To further make a fair comparison, we reduce the effect of different backbones by comparing the performance ratio between each method and its fully-supervised counterpart on the VOC validation. As shown in Tab. 3, our method and CTI [31] achieve the highest performance ratio with the same 91.7%, outperforming other WSSS methods. For example, compared to DuPL [13], we achieve a 2.6% gain of performance ratio. Despite the same performance ratio between our method and CTI, we significantly surpass CTI by 1.4% mIoU.

Visualization Results. We first conduct the visualization comparison of segmentation masks generated by our method, ToCo [12], DuPL [13] and the ground truth, as shown in Fig. 4. It can be observed that our method effectively mitigates the issue of over-expansion, producing more accurate and complete segmentation masks that are close to the ground truth.

Furthermore, we visualize the CAM on PASCAL VOC 2012 to compare our method with SOTA one-stage WSSS methods, as shown in Fig. 6.

4.3. Ablation studies and Analysis

Effectiveness of Components. We conduct ablation experiments on the VOC 2012 validation set to assess each module. Following [12, 13], we use ViT-B/16 as our baseline. The experimental results are presented in Tab. 4. First, **Patch-level Confidence Allocation** (PCA), distributes confidence between classes and patches, enabling more precise localization of high-confidence regions, thereby improving segmentation performance. Second, the **Progressive Mask Expansion and Combination** (PMEC) strategy gradually expands the mask from high-confidence patches to the entire target, effectively suppressing interference from similar regions. This approach increases segmentation accuracy to 72.5% mIoU, yielding a 2% improvement over using PCA alone. Based on CRME, the $\mathcal{L}_{\text{CPE}}$ loss significantly enhances the alignment of features and class tokens across different classes within the semantic space, further improving segmentation accuracy by 0.7% mIoU. Finally, the **Progressive Class Prototype Update** (PCPU), dynamically refines class prototypes during training, which achieves segmentation performance improvement of 0.6% mIoU compared to CRME+$\mathcal{L}_{\text{CPE}}$.

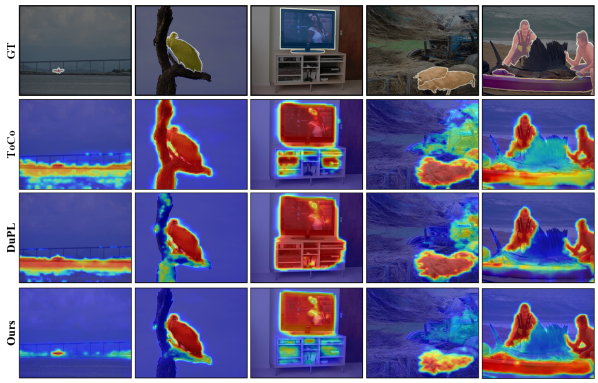


Figure 6. **Visualization results of CAM on PASCAL VOC 2012.** Our method produces more accurate and complete CAMs compared to other methods.

$\mathcal{L}_{\text{conf}}$	Confidence Dimension	CAM	Seg.
Max	Single	71.5	69.4
Sum	Single	73.1	71.0
Max	Patch-Class	74.5	73.0
Sum	**Patch-Class**	**76.3**	**73.8**

$\tau_{\text{low}} \backslash \tau_{\text{high}}$	0.9	0.7	0.5
0.4	73.0	73.4	**73.8**
0.3	72.4	72.6	73.2
0.2	71.9	72.3	72.6

Table 5. **Ablation studies.** *Left*: Patch-Level Confidence Allocation configurations. *Right*: ROR thresholds on VOC validation set (mIoU%).

Further Analysis of CRME. We first analyze the effects of confidence allocation configurations in CRME, *i.e.*, patch-class joint confidence *vs.* single patch-wise confidence. Additionally, to refine the supervision strategy, we compare the effectiveness of sum loss and max loss in Eq. 4. As shown in Tab. 5 *Left*, patch-class joint confidence significantly enhances segmentation performance, demonstrating the advantage of jointly modeling class and spatial information. Furthermore, the sum loss outperforms the max loss in overall performance. Second, we conduct the ablation study on ROR thresholds, as shown in Tab. 5 *Right*. τ_{low} and τ_{high} are set to 0.4 and 0.5, which achieves the best segmentation performance.

5. Conclusion

In this paper, we propose a Progressive Confidence Region Expansion (PCRE) framework to address the *over-expansion* issue in CAM for weakly supervised semantic segmentation tasks. PCRE allows the gradual generalization of a faithful mask covering the entire target region and introduces representative, dataset-level prototypes for each class, which rectify excessive expansion to similar background areas in CAM. Extensive experiments show that PCRE achieves new state-of-the-art performances.

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